

Track B: Detailed Methods and Results

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This document expands the compact Track B handoff into a full methods-and-results reference suitable for a paper appendix: it defines every method and metric, states the exact computations, and reports all result tables gathered so far. Track B is the set of eight follow-up experiments (B1–B8) requested in the TMLR revision to strengthen the TraLO paper. *TraLO* trains a network to satisfy a **transductive prediction-count constraint** (an upper bound on how many test points may be predicted as a designated “constrained” class) by adding a soft, bounded penalty to the training loss, and is compared against constrained-optimization duals, post-hoc clipping, and imbalanced-learning baselines.

Status. B1–B8 are complete except B1’s third backbone (ViT-B/16), which is still computing; its rows are marked *pending* in Table 1 and do not affect any stated conclusion.

1 Problem setup and notation

We are given a trained-on set and an unlabeled *test* (transductive) set. One class c^* is designated *constrained*. A deployment rule caps the number of test points predicted as c^* :

$$\underbrace{\sum_{i \in \text{test}} \mathbf{1}[\hat{y}_i = c^*]}_{\text{hard count}} \leq K, \quad K = (\text{cap fraction}) \times (\text{true prevalence of } c^*). \quad (1)$$

The **soft count** $S_{c^*} = \sum_i p_i(c^*)$ (sum of softmax probabilities) is the differentiable surrogate used inside the loss; the **hard count** (using $\arg \max$) is used only to *verify* satisfaction. The *excess* is $E = \max(0, S_{c^*} - K)$. Caps come in two flavours: a *global* cap G on the total count and *local* caps L on per-group counts; a cell tagged L30_G30 sets both to 30% of the true prevalence (symmetric). Smaller percentages are *tighter* caps and bind harder.

2 Methods compared

All methods produce a classifier and then must meet the cap. They differ in *how*: TraLO and the duals *train* to satisfy it; the imbalanced baselines train an imbalance-aware classifier and then clip; post-hoc clipping trains normally and only edits predictions at inference.

2.1 TraLO (proposed)

Two phases. (i) **Warm-up**: standard cross-entropy (CE) for W epochs. (ii) **Constraint phase** (up to 300 epochs): minimise

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_g \mathcal{L}_{\text{global}} + \lambda_\ell \mathcal{L}_{\text{local}}, \quad \mathcal{L}_{\text{pen}} = \underbrace{\frac{E}{E+K}}_{\text{rational saturation}} + \underbrace{\rho \frac{(E/K)^2}{1+(E/K)^2}}_{\text{bounded quadratic}} \in [0, 1+\rho). \quad (2)$$

The penalty is deliberately **bounded** (it saturates as the violation grows), unlike the duals’ linear penalty. Training uses several controls, each individually ablated in the paper: a *λ -ratchet* (increment λ each epoch until the cap is met, then freeze); a *linear ρ schedule* (raise ρ until first satisfaction, then freeze); a *hinge / λ -toggle* (zero the multipliers when satisfied, restore when violated, with oscillation detection); an *optimizer reset at first satisfaction* (reset the Adam moment estimates to kill post-satisfaction momentum); a *CE-saturation skip* (end phase ii when train accuracy ≥ 0.995); and *convergence early-stopping* (5 consecutive satisfied epochs).

2.2 Constrained-optimization duals

These solve the constrained problem with a Lagrange multiplier λ updated by dual ascent.

- **Fioretto-LDF** (linear dual): penalty $\lambda(S_{c^*} - K)$ with $\lambda \leftarrow [\lambda + \eta(S_{c^*} - K)]_+$. The linear penalty can *wind up* (λ grows without bound if the cap is never met).
- **ALM** (augmented Lagrangian): $\lambda \leftarrow \max(0, \lambda + \eta(S_{c^*} - K)) + \mu(S_{c^*} - K)^+$ with μ growing linearly. This is the standard literature fix for linear-penalty windup, and empirically a *stronger* baseline than plain Fioretto.
- **Hounie-RCL** (resilient constrained learning): allows the constraint level to relax adaptively at a cost. In the tight-cap regime it underperforms sharply (see B8).

2.3 Post-hoc clipping (Shifman-LP)

Train the network with plain CE (no constraint), then at inference solve a small linear program that flips the minimal set of borderline predictions so the hard count meets K . It is **post-hoc**: it never trains to satisfy the cap and therefore *cannot re-rank* the decision boundary; it only relabels the least-confident predictions. It is a different category of method from the trained ones, and is reported separately.

2.4 Imbalanced-learning baselines (B1)

These target the rare constrained class directly. Each trains the backbone with an imbalance-aware loss during the warm-up phase and then applies Shifman-LP clipping; the warm-up is therefore their *entire* training budget. Let p_y be the softmax probability of the true class y , n_c the number of training samples in class c , and π_c the class prior (frequency).

- **Focal loss** (Lin et al., 2017): $\ell_{\text{focal}} = -\alpha(1 - p_y)^\gamma \log p_y$, with $\alpha = 0.25$, $\gamma = 2.0$. The modulating factor $(1 - p_y)^\gamma$ down-weights easy (high-confidence) examples, concentrating gradient on hard / rare cases and boosting minority-class recall.
- **Class-balanced weighting** (Cui et al., 2019): weight class c 's loss by the inverse *effective number* of samples, $w_c = \frac{1 - \beta}{1 - \beta^{n_c}}$, with $\beta = 0.9999$. The effective number $(1 - \beta^{n_c}) / (1 - \beta)$ discounts information overlap among many similar samples, so rare classes receive proportionally larger weight.
- **Logit adjustment** (Menon et al., 2020): add a prior-based offset to the logits, $z_c \leftarrow z_c + \tau \log \pi_c$, with $\tau = 1.0$. This shifts the decision threshold to compensate for label imbalance and is Bayes-consistent for the balanced error rate.

3 Metrics, columns, and parameters

- **cc-F1 (constrained-class F1)** – the F1 score of the constrained class c^* alone; the primary deployment metric. Because a cap at fraction c of the true count limits recall to $\leq c$, the attainable cc-F1 is *ceiling-bounded* by $2c/(1 + c)$ (e.g. $c = 0.3 \Rightarrow 0.46$). Absolute cc-F1 gaps are therefore small by construction (“ceiling crush”).
- **macro-F1** – the unweighted mean of per-class F1; overall classification quality across all classes. It is the metric the imbalanced baselines (B1) target.
- **Native satisfaction** – the fraction of runs whose hard (argmax) count already meets the cap at inference *without* any post-hoc flip. Methods that train to satisfy (TraLO, duals) reach 1.0; methods that clip afterwards (post-hoc clip, imbalanced baselines) are 0.0 and rely on the LP.

- **Cap tag** LX_GX – local and global caps each set to $X\%$ of the true prevalence of c^* ; smaller X = tighter.
- **Warm-up epochs** W – CE-only pre-training before the constraint phase, and the “headroom” lever: short W leaves CE unsaturated (train-acc < 1) so the penalty can re-rank the boundary; long W saturates CE (train-acc $\rightarrow 1$), leaving only a uniform logit shift.
- **Gap** “ $A - B$ ” – the matched-seed mean of $(\text{metric}_A - \text{metric}_B)$ within a cell; positive means A is better.
- **Cell** – the atomic unit of analysis: one (dataset, backbone, cap, method) averaged over the 4 seeds. Backbones, datasets, and caps are **never pooled**.
- **W/T/L at τ** – across cells, the counts with gap $\geq +\tau$ (win), $|\text{gap}| < \tau$ (tie), and $\leq -\tau$ (loss).

4 Statistical methodology (the computations)

- **Matched-seed (paired) comparison.** A seed fixes the initialization, data order, and warm-up, so two methods at the same seed start from the same network; we compare their per-seed difference. This pairing removes between-seed variance, so we *never* use pooled-standard-deviation (unpaired) tests.
- **Bootstrap confidence intervals.** For a comparison at a given cap, we resample the constituent cells (backbones) with replacement 20,000 times and take the 2.5/97.5 percentiles of the resampled mean gap. A CI that excludes 0 is significant.
- **Benjamini–Hochberg FDR.** Across the family of per-cell paired tests we control the false discovery rate at $q = 0.05$ and report how many cells survive.
- **Win-bar rule.** A regime counts as a “win” if the mean paired cc-F1 gap $\geq \tau$ *and* at least half the cells individually clear τ ; we sweep $\tau \in \{0.003, 0.005, 0.010\}$ to show the classification is not threshold-sensitive.

Every experiment ran 4 seeds (1–4); the campaign totals 568 runs (plus 60 ViT-B/16 runs for B1 in progress), with 0 failures.

5 Experiments and results

5.1 B1 – Imbalanced-learning baselines

Question. Does an imbalance-aware clipper (focal / class-balanced / logit-adjust + clipping) close the macro-F1 gap the paper attributes to constraint-time training? **What ran.** 252 runs: the 5 methods $\times \{\text{Oct, Derm, Tissue}\}\text{MNIST} \times \{\text{MobileNetV3, RegNetY-400MF}\} \times \text{warm-up } \{1, 5, 50\} \times \text{cap L30} \times 4 \text{ seeds}$. Table 1 reports the macro-F1 gap (TraLO – baseline) at warm-up 50. **Found.** Only focal is competitive; it *beats* TraLO on Derm/MNV3 and *loses* on Oct/MNV3, i.e. the outcome is backbone- and dataset-dependent. cc-F1 is tied (ceiling). The durable, universal separation is native satisfaction (1.0 vs 0.0). *Verdict: an honest tie on macro-F1, plus native satisfaction.*

5.2 B2 – Native-resolution replication

Question. Does the tight-cap behaviour reproduce at native (224px) resolution? **What ran.** 192 runs on five native MedMNIST datasets (Derm=HAM10000, Retina, Blood, OrganA, TissueNative) $\times \text{warm-up } \{1, 5\} \times \text{cap L30 (+L20)} \times 4 \text{ seeds}$. We compare against the best *trained* baseline

dataset	backbone	vs focal	vs class-bal	vs logit-adj	vs clip
Derm	MobileNetV3	−0.012	−0.016	+0.021	+0.010
Derm	RegNetY400MF	+0.006	+0.022	+0.042	+0.013
Oct	MobileNetV3	+0.025	+0.023	+0.021	+0.023
Oct	RegNetY400MF	−0.005	+0.022	−0.000	+0.022
Tissue	MobileNetV3	+0.020	+0.014	+0.026	+0.029
Tissue	RegNetY400MF	−0.002	+0.003	−0.001	+0.001
<i>ViT-B/16 (third backbone): runs in progress; rows to be appended.</i>					

Table 1: B1: macro-F1 gap (TraLO − baseline) at warm-up 50, cap L30, mean over 4 seeds. Positive = TraLO better. Focal is the only competitive baseline and is backbone-dependent.

(Fioretto-LDF); post-hoc clip is excluded here because at warm-up 1/5 it is undertrained, so a gap over it would measure “clip did not train,” not a real advantage; Hounie was not run natively. **Found** (Table 2): TraLO *ties* the best trained baseline on all five datasets and both warm-ups ($|\Delta| \leq 0.011$) – an honest negative replication, consistent with the mechanism (both trained methods re-rank equally when CE has headroom). Native satisfaction remains 1.0.

dataset	TraLO (w1 / w5)	best baseline, Fioretto (w1 / w5)	$\Delta = \text{TraLO} - \text{best}$ (w1 / w5)
Derm	0.331 / 0.412	0.325 / 0.412	+0.006 / −0.000
Retina	0.181 / 0.208	0.181 / 0.200	+0.000 / +0.008
TissueNat	0.212 / 0.265	0.211 / 0.254	+0.001 / +0.011
OrganA	0.395 / 0.463	0.392 / 0.463	+0.003 / +0.000
Blood	0.460 / 0.457	0.460 / 0.457	+0.000 / +0.000

Table 2: B2: cc-F1 at native 224px, TraLO vs the best trained baseline (Fioretto-LDF), each cell warm-up 1/5, mean over 4 seeds. A tie across all datasets and warm-ups.

5.3 B3 – ALM (augmented-Lagrangian) baseline

Question. Does ALM, the standard fix for linear-penalty windup, close the gap? **What ran.** 24 runs (Fioretto-LDF with the ALM update) on OctMNIST L30/L40 $\times$ {MNV3, RegNet, ViT} $\times$ 4 seeds, adjudicated against the frozen TraLO and Fioretto cells. **Found** (Table 3): ALM is a *stronger* dual than plain Fioretto (it beats it by +0.010), yet TraLO tops ALM in all 6 tight-cap cells (+0.028 cc-F1). ALM does not close the gap, so the paper’s justification for not adopting ALM stands. This is the strongest new result.

comparison (cc-F1)	mean	W/T/L	MNV3 (L30/L40)	RegNet (L30/L40)	ViT (L30/L40)
TraLO − ALM (best trained base.)	+0.028	6/0/0	+0.009 / +0.013	+0.048 / +0.008	+0.046 / +0.046
TraLO − Fioretto	+0.039	6/0/0	+0.016 / +0.022	+0.035 / +0.021	+0.086 / +0.051
ALM − Fioretto	+0.010	5/0/1	+0.007 / +0.009	−0.013 / +0.013	+0.041 / +0.005

Table 3: B3: OctMNIST cc-F1 gaps, mean over 4 seeds; 6 cells each. Raw cc-F1: TraLO 0.455 > ALM 0.427 > Fioretto 0.417.

5.4 B4 – Tie-region component ablation

Question. Are the TraLO controls (reset / hinge / ρ -schedule / freeze) load-bearing in a *tie* region, or only at tight caps? **What ran.** 20 runs: DermMNIST L50 (a tie cell), MNV3, leave-one-out of

each component $\times 4$ seeds. **Found** (Table 4): removing any single component moves results by at most ± 0.006 – none is load-bearing in a tie region. This is the tighter (and favourable) conclusion: the portable components carry the *tight-cap* advantage specifically, consistent with the mechanism.

removed	macro-F1 Δ	cc-F1 Δ	removed	macro-F1 Δ	cc-F1 Δ
– reset	+0.002	+0.006	– ρ -schedule	+0.002	+0.003
– hinge	–0.003	0.000	– freeze	–0.002	+0.003

Table 4: B4: leave-one-out deltas (full – variant) on the DermMNIST L50 tie cell, mean over 4 seeds. Full run: 0.744 macro-F1 / 0.564 cc-F1. No component is load-bearing here.

5.5 B5 – Win-bar sensitivity

Question. Is the OctMNIST regime classification stable across win thresholds τ ? **What ran.** No new runs; the OctMNIST cc-F1 cells were re-scored at $\tau \in \{.003, .005, .010\}$. **Found** (Table 5): the win vs the trained baselines at L30/L40 is stable at every threshold (4/0/0 vs Fioretto even at the strict $\tau = .010$), while the comparison against post-hoc clip never yields a stable win at any threshold – the honest cc-F1 tie.

vs. trained base.	cap	.003	.005	.010	vs. clip (separate)	cap	.003	.005	.010
TraLO – Fioretto	L30	4/0/0	4/0/0	4/0/0	TraLO – clip	L40	3/0/1	2/1/1	1/2/1
	L40	4/0/0	4/0/0	4/0/0		L30	2/0/2	2/0/2	0/2/2
TraLO – ALM	L30	3/0/0	3/0/0	2/1/0		L10–20	tie	tie	tie

Table 5: B5: W/T/L across OctMNIST backbone cells by cap and win threshold τ . Left: vs the trained baselines (duals). Right: vs post-hoc clip (separate category).

5.6 B6 – Bootstrap confidence intervals; B7 – BH-FDR

Question. Do the headline cells survive $n = 4$ seeds statistically? **What ran.** No new runs; 20,000-resample bootstrap CIs over the OctMNIST backbone cells per cap, plus BH-FDR. **Found** (Table 6): CIs against the trained baselines *exclude* 0 at L30/L40 (the previously-flagged ViT L30 cell is +0.086 vs Fioretto and survives), while the CI against post-hoc clip *includes* 0 at every cap. The win is a mid-cap (L30/L40) phenomenon: at L10/L15/L20 even the dual gap ties because the tightest caps ceiling-crush every method. **B7 (BH-FDR, $q = 0.05$):** 3/8 cells survive for TraLO – Fioretto; 0/8 for TraLO – clip.

vs. trained	cap	mean	95% CI	vs. clip (separate)	cap	mean	95% CI
TraLO – Fioretto	L30	+0.046	[+.024, +.074]	TraLO – clip	L10	+0.001	[–.005, +.007]
	L40	+0.033	[+.021, +.044]		L15	+0.004	[.000, +.007]
	L20	+0.002	[.000, +.005]		L20	+0.001	[–.008, +.007]
TraLO – ALM	L30	+0.035	[+.010, +.048]		L30	–0.003	[–.013, +.008]
	L40	+0.022	[+.008, +.046]		L40	+0.002	[–.010, +.012]

Table 6: B6: grand-mean OctMNIST cc-F1 gap with 95% bootstrap CI (resampling backbone cells). Trained-baseline CIs exclude 0 at L30/L40; the clip CI includes 0 at every cap.

5.7 B8 – Fourth backbone (MobileNetV2)

Question. Does the cc-F1 win reproduce on a fourth backbone? **What ran.** 32 runs: MobileNetV2 $\times$ OctMNIST L30/L40 $\times$ {TraLO, Fioretto, Hounie, clip} $\times$ 4 seeds. **Found** (Table 7): TraLO beats the best trained baseline (Fioretto) by +0.042 and Hounie collapses; the comparison vs post-hoc clip is a thin cc-F1 tie (+0.007) with 0 native satisfaction. The dual win is thus backbone-general across MNV3/RegNet/ViT/MNV2.

comparison (cc-F1)	mean	L30	L40	raw cc-F1 (MNV2)
TraLO – Fioretto (best trained base.)	+0.042	+0.047	+0.037	TraLO 0.454
TraLO – Hounie (collapses)	+0.169	+0.189	+0.149	Fioretto 0.412
TraLO – clip (separate category)	+0.007	+0.009	+0.005	Hounie 0.285, clip 0.447

Table 7: B8: MobileNetV2, OctMNIST cc-F1 gaps, mean over 4 seeds. The dual win reproduces on a fourth backbone.

6 What worked, what did not, and what was researched

What worked (goes into the paper).

- TraLO beats the *best trained baseline* on OctMNIST tight caps: +0.028 cc-F1 vs the strong ALM and +0.039 vs Fioretto (6/6 cells), with bootstrap CIs excluding 0 at L30/L40 (B3, B6) and stable under the win-bar sweep and BH-FDR (B5, B7).
- The win is **backbone-general**: it reproduces on MNV3, RegNet, ViT, and MNV2 (B8).
- The controls are **mechanism-consistent**: they are load-bearing at tight caps but inert in a tie region (B4), which is exactly what the mechanism predicts.
- TraLO uniquely provides **native constraint satisfaction** (1.0) across every setting.

What did not work / honest negatives (reported as scope).

- Against **post-hoc clipping**, TraLO’s cc-F1 is a *tie*, not a win: the gap CI includes 0 at every cap (B5, B6). Clipping matches accuracy but does not satisfy the cap natively, so the honest claim vs clip is “parity + native satisfaction,” not an accuracy win.
- At **native resolution** the tight-cap advantage does *not* appear: TraLO ties the best trained baseline on all five datasets (B2). An earlier apparent TissueNative edge did not survive adding a backbone and a tighter cap.
- **Focal loss** matches or beats TraLO’s macro-F1 on DermMNIST/MNV3 (B1); the macro-F1 story is backbone-dependent, not a clean TraLO win.

What was researched (the scientific contribution). The *headroom mechanism* (Section 7) — a precise characterization of *when* transductive count-constraint training helps — which unifies every win and every tie above.

7 The mechanism

The count penalty (2) reaches the network weights only through the scalar soft-count S_{c^*} , so its gradient is a *scalar $\times$ a fixed direction* and its overall scale (λ, ρ) is absorbed by the Adam optimizer and gradient clipping. What the penalty can *do* therefore depends entirely on whether CE is still active:

- **CE active** (tight cap + warm-up headroom): the penalty *re-ranks* borderline examples, producing a genuine cc-F1 gain over the trained duals (B3, B8).
- **CE saturated** (loose cap, long warm-up, easy data, or native resolution): the penalty can only *shift* logits uniformly, leaving the top- K set — and thus the metric — unchanged, giving a tie (B2, B4, B5–B6).

Post-hoc clipping sits on a separate axis: it can match cc-F1 but, being applied after training, can never satisfy the cap natively. This single mechanism predicts every Track B win and tie, and is supported by the Adam scale-invariance analysis in the main text.

8 Limitations and honest scope

The cc-F1 advantage is over the trained *duals*, and only at mid tight caps (L30/L40); it narrows to a tie at the very tightest caps (L10–L20), where every method is ceiling-crushed. Against post-hoc clipping it is a tie plus native satisfaction. Focal ties TraLO on macro-F1 on DermMNIST. Native resolution is a trained-baseline tie. cc-F1 gaps are small by construction (ceiling $2c/(1+c)$). Framed as a characterization of *when* count-constraint training helps — beating the trained duals where the cap binds and the network is unsaturated, matching strong baselines elsewhere, and uniquely satisfying the cap natively — this is a precise and defensible contribution.